

# **Break Even Analysis and Short term profit planning**

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# Break-Even Analysis

- Break-Even Analysis is used to
    - predict future profits/losses
    - predict results eg produce Product A or Product B
  - Break-Even Point is when Sales Revenue equals Total Costs at this point no profit or loss is incurred the firm merely covers its total costs
  - Break-Even Point can be shown in graph form or by use of formulae
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# Break-Even Analysis

There are two basic types of costs a company incurs.

- **Variable Costs**
  - **Fixed Costs**
- **Variable costs** are costs that change with changes in production levels or sales. Examples include: Costs of materials used in the production of the goods, production labour (hourly or piecework wages), sales commissions, repairs, maintenance, and more. This is also referred as **direct cost** of the company. Direct material, direct labor, direct variable cost.
- **Fixed costs** remain roughly the same regardless of sales/output levels. Examples include: rent, insurance, property taxes, salaries unrelated to production (such as management), production equipment, office furniture, and much more.
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# Break-Even Analysis

- TOTAL COSTS

- Total Costs is simply Fixed Costs and Variable Costs added together.

$$TC = FC + VC$$

- As Total Costs include some of the Variable Costs then Total Costs will also change with any changes in output/sales.
- If output/sales rise then so will Total Costs.
- If output/sales fall then so will Total Costs.



# Break-Even Analysis

The Break-even point occurs when Total Costs equals Revenue (Sales Income)

$$\text{Revenues (Sales Income)} = \text{Total Costs}$$

At this point the business is not making a **Profit** nor incurring a **Loss** – it is merely covering its Total Costs

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# Graphical Method

$$TR = P.Q$$

As, **P** is fixed

$$TR = P.Q \quad \square \quad TR = f(Q)$$

this tells that revenue, profit and total cost depends on quantity

$$\text{As, } TC = FC + VC$$

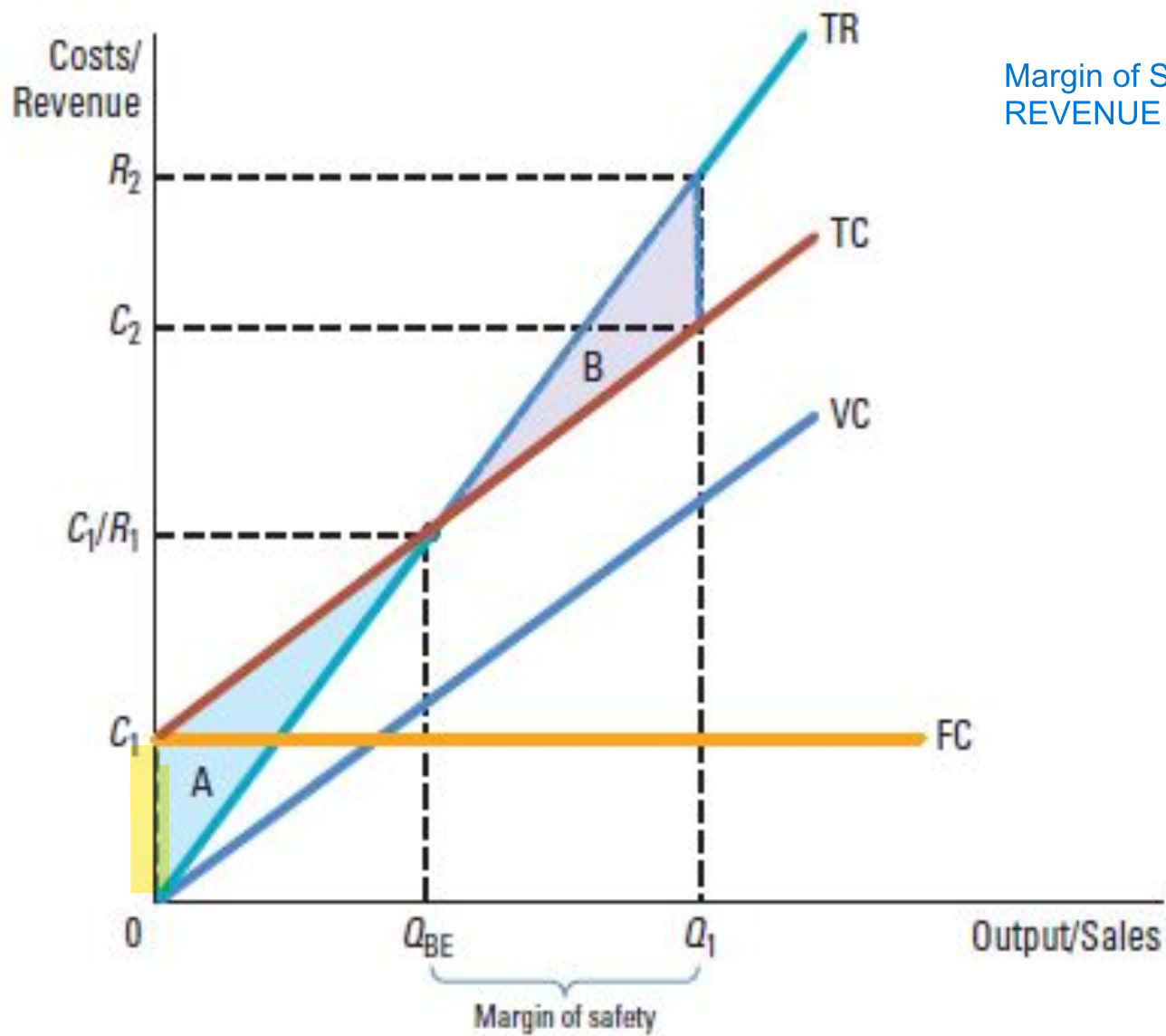
$$TC = FC + AVC(Q) \implies TC = c(Q)$$

$$\text{As, } \Pi = TR - TC$$

$$\Pi = f(Q) - c(Q)$$

$$\Pi = \Pi(Q)$$

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Margin of Safety (MOS) = TOTAL REVENUE - BREAK EVEN SALES

# Break Even Analysis - Total Contribution

Break Even Output ( $Q^*$ ) is the output level where Total Revenue is equal to Total Cost or Profit and loss is zero i.e.

$$TR = TC \text{ or } \pi = 0$$

As, Profit = Total Revenue – Total Cost

$$\pi = TR - TC$$

$$\pi = TR - FC - VC \quad [\text{As, } TC = FC + VC]$$

$$TR - VC = \pi + FC \text{ ----- (1)}$$

$$\text{Total Contribution} = \pi + FC$$

[Total contribution of sales is used to cover Fixed Cost (FC) and earn profit]

$$\text{Eq. (1)} \Rightarrow TR - VC = FC \quad [\text{As, At Break Even, } \pi = 0]$$

At break even, Total contribution (TR-VC) is just able to cover the Fixed Cost (FC).

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# Break Even Analysis - Unit Contribution Or Contribution Margin

$$\text{Total contribution} = \text{TR} - \text{VC}$$

$$\text{Total Contribution} = \text{P.Q} - \text{AVC.Q}$$

Divide both sides by output produced (Q),

$$\text{Total contribution}/\text{Q} = \text{TR}/\text{Q} - \text{VC}/\text{Q}$$

$$\text{Unit Contribution} = \text{P.Q}/\text{Q} - \text{AVC.Q} / \text{Q}$$

$$\text{Unit Contribution} = \text{P} - \text{AVC} \quad \text{OR}$$

$$\text{Unit Contribution} = \text{AR} - \text{AVC}$$

# Break Even Output

As, at Break even,  $TR - VC = FC$  ----- (2)

Define contribution in terms of units produced,  
i.e., Eq. (2)  $\Rightarrow P.Q - AVC.Q = FC$

$$Q (P - AVC) = FC$$

$$Q^* = FC / (P - AVC) \text{ Break Even Output}$$

# Break-Even Analysis

Let us have a look at a simple example. Bannerman Trading Company opens a flower shop.

## Fixed Costs:

- Rent: Rs.400
- Helper (Wages): Rs.200

## Variable Costs:

- Flowers: Rs.0.5 per unit

## Selling Price:

- Flowers: Rs.2 per unit

So we know that:

Total **Fixed Costs** = **Rs.600**

**Variable Cost** per Unit = **Rs.0.50**

**Selling Price** per Unit = **Rs.2.00**



# Break-Even Analysis

$$SP = \text{£}2.00$$

$$VC = \text{£}0.50$$

$$FC = \text{£}600$$

We must firstly calculate how much income from each bunch of flowers can go towards covering the **Fixed Costs**. This is called the **Unit Contribution**.

Selling Price – Variable Costs = Unit Contribution

$$\text{Rs.}2.00 - \text{Rs.}0.50 = \text{Rs.}1.50$$



- For every bunch of flowers sold **Rs.1.50** can go towards covering **Fixed Costs**

# Break-Even Analysis

SP = Rs.2.00

VC = Rs.0.50

Unit CPU =  
Rs.1.50

FC = Rs.600

Now to calculate how many units must  
be sold to cover **Total Costs** (FC + VC)

This is called the **Break Even Point**

**Break Even Point** = **Fixed Costs** ÷ **Unit Contribution**

$$\text{Rs.600} \div \text{Rs.1.50} = 400 \text{ Units}$$



Therefore 400 bunches of flowers must be sold to **Break Even** – at this the point the business is not making a **Profit** nor incurring a **Loss** – it is merely covering its **Total Costs**

# Break-Even Analysis

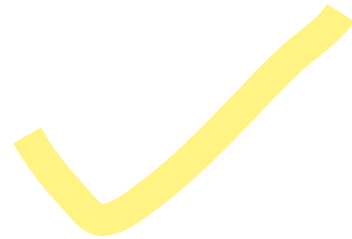
Lets try another example:

Selling Price per unit = Rs.5

Variable Cost per unit = Rs.2

Fixed Costs = Rs.300

How many units must be sold in order to Break Even?



# Break-Even Analysis

$$SP = \text{£}5.00$$

$$VC = \text{£}2.00$$

$$FC = \text{£}300$$

First calculate the Unit Contribution

$$SP - VC = \text{Unit Contribution}$$

$$\text{£}5.00 - \text{£}2.00 = \text{£}3.00$$

Now calculate Break Even point by using the formula –

$$\text{Fixed Costs} \div \text{Unit Contribution}$$

$$\text{£}300 \div \text{£}3.00 = 100 \text{ units}$$

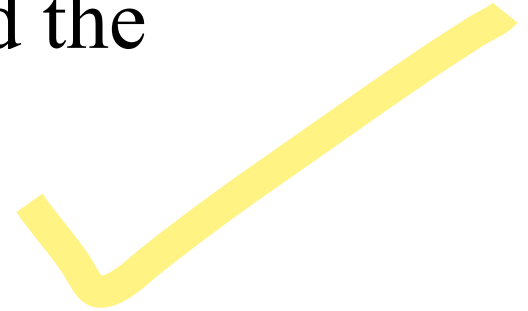
Therefore 100 units must be sold in order to Break Even

# Break-Even Analysis

Lets try another example:

A firm has Fixed Costs of £1,200.

The Selling Price is £6 per unit and the  
Variable Costs are £3 per unit.



How many units must be sold in order to Break  
Even?



# Break-Even Analysis

$$SP = \text{£}6.00$$

$$VC = \text{£}3.00$$

$$FC = \text{£}1,200$$

First calculate the Unit Contribution

$$SP - VC = \text{Unit Contribution}$$

$$\text{£}6.00 - \text{£}3.00 = \text{£}3.00$$

Now calculate Break Even point by using the formula –

$$\text{Fixed Costs} \div \text{Unit Contribution}$$

$$\text{£}1,200 \div \text{£}3.00 = 400 \text{ units}$$

Therefore 400 units must be sold in order to Break Even



# Break Even Sales (BES)

$$Q^* = FC / (P - AVC)$$

$$P.Q^* = FC (P) / (P - AVC) \text{ [Multiplied both sides by Price (P)]}$$

$$P.Q^* = FC (P)(Q) / (P - AVC)(Q) \text{ [Right side, Multiplied and divided by output 'Q']}$$

$$BES = FC * TR / \text{Total Contribution}$$

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# Break-Even Analysis

**Fixed Costs =**

**Rent                Rs.1,050**

**Insurance        Rs. 200**

**Total FC        Rs.1,250**

Caroline Wilson owns a florist shop.

She buys each bunch of flowers for **Rs.1.49** and special wrapping paper for **Rs.3 per roll**. Each roll of wrapping paper will wrap **300 bunches of flowers**. Rent of her premises is **Rs.1,050 per month** and she pays monthly insurance of **Rs.200**. Caroline sells each bunch of flowers for **Rs.2.50**.

Calculate the Break even point and revenue 11. Monthly profitability

# Break-Even Analysis

**1. Identify fixed costs** Fixed costs (FC) are the monthly overhead expenses that do not change based on the number of flowers sold.

Rent: Rs.1,050

Insurance: Rs.200

Total monthly fixed costs (FC):  $1050 + 200 = \text{Rs.}1250$

**2. Identify variable costs** Variable costs (VC) are the expenses incurred for every bunch of flowers sold. This includes the cost of the flowers and the portion of wrapping paper used per bunch.

Cost of one bunch of flowers: Rs. 1.49

Cost of wrapping paper per bunch:

$\text{Rs } 3 / 300 = \text{Rs } 0.01/\text{bunch}$

Total variable cost per unit (v):  $1.49 + 0.01 = \text{Rs.}1.5$



# Break-Even Analysis

## 3. Calculate contribution margin:

Selling price (p): Rs.2.50

Contribution margin (CM):  $P - AVC = 2.5 - 1.5 = \text{Rs.}1$

## 4. Determine break-even point

$$Q^* = FC / CM$$

$$Q^* = 1,250 / 1.00 = 1,250 \text{ bunches}$$



The break-even revenue is the total sales required to cover all costs:

$$\text{Revenue} = P * Q = 2.50 * 1,250 = \text{Rs.}3,125$$

# Break-Even Analysis

## 5. Analyze monthly profitability

Profitability depends on the volume sold above the break-even point. For every bunch sold beyond 1,250, Caroline earns a profit of Rs. 1.00 (the contribution margin).

If sales < 1,250: The business operates at a loss.

If sales = 1,250: The business breaks even (zero profit).

If sales > 1,250: Profit = (Units sold - 1, 250) × Rs.1.00.

## Break-Even Analysis and Target profit concept

Calculate how many units need to be produced in order to achieve a Net Profit of £25,000 given the following information

Fixed Costs Rs.30,000. Selling price per unit Rs.50 and VC per unit Rs.40.00



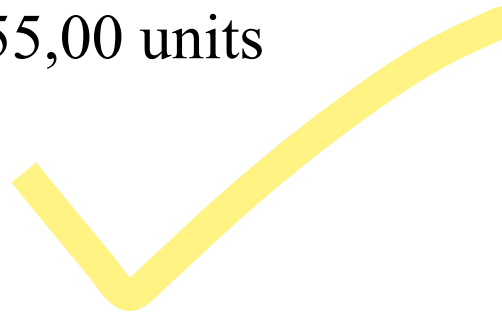
# Answer

Number of units  
required to earn a  
target profit

$$= [\text{Fixed cost} + \text{Target Profit}] / \text{CPU}$$

$$= 30,000 + 25,000 / 10.00$$

$$= 55,00 \text{ units}$$





# Break-Even Analysis

The formulae used **so far assumes that** Unit Costs are known i.e. Unit Selling Price and Unit Variable Cost.

**When no unit costs are known, the Profit/Volume Ratio should be used instead**

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# Profit Volume Ratio OR Contribution Sales Ratio (P/V Ratio)

$P/V \text{ Ratio} = \text{Total Contribution} / \text{Total Revenue}$

$P/V \text{ Ratio} = (TR - VC) / \text{Sales}$

$P/V \text{ Ratio} = TR / \text{Sales} - VC / \text{Sales}$

$P/V \text{ Ratio} = 1 - VC / \text{Sales}$

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# Break Even Sales (BES)

## Finding Break Even sales via P/V Ratio

$$\text{BES} = \text{FC} * \text{TR} / \text{Total Contribution}$$

$$(\text{BES}) = \text{FC} * \text{TR} / \text{Sales} \div \text{Total Contribution} / \text{Sales}$$

Right side divided and multiplied by sales]

$$(\text{BES}) = \text{FC} / (\text{P/V Ratio})$$

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# Profit/Volume Ratio

## For Example

Selling price per unit Rs.80.00

Variable cost per unit Rs.40.00

Fixed Costs Rs.120,000

Calculate the C/S Ratio and the BEP value

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# Assumptions of Break Even Analysis

- All Fixed and Variable costs can be identified
  - Variable costs are assumed to vary directly with output
  - Fixed costs will remain constant
  - Selling prices are assumed to remain constant for all levels of output
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# Assumptions Continued

- The sales mix of products will remain constant – break even charts cannot handle multi-product situations
  - It is assumed that all production will be sold
  - The volume of activity is the only relevant factor which will affect costs
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# Limitations of Break Even Analysis

- Some costs cannot be identified as precisely Fixed or Variable
  - Semi-variable costs cannot be easily accommodated in break-even analysis
  - Costs and revenues tend not to be constant
  - With Fixed costs the assumption that they are constant over the whole range of output from zero to maximum capacity is unrealistic
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# Limitations Continued

- Price reduction may be necessary to protect sales in the face of increased competition
  - The sales mix may change with changes in tastes and fashions
  - Productivity may be affected by strikes and absenteeism
  - The balance between Fixed and Variable costs may be altered by new technology
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# Break-Even Analysis and Margin of safety

Break Even can also be used to calculate Profit (or Loss) at a given level of output

For example:

J Bannerman sells cricket bats. How much profit/loss is made when 5000 cricket bats are sold?

Each cricket bat is sold for Rs.20

Variable Costs per cricket bat are Rs.10

Fixed Costs total Rs.24,000



# Break-Even Analysis



$$SP = \text{Rs.}20.00$$

$$VC = \text{Rs.}10.00$$

$$FC = \text{Rs.}24,000$$

$$\text{Sales} = 5,000 \text{ units}$$

Firstly, calculate Unit Contribution

$$SP - VC = \text{Unit Contribution}$$

$$\text{Rs.}20.00 - \text{Rs.}10.00 = \text{Rs.}10.00$$

Now calculate Total Contribution when 5,000 golf clubs are sold

$$\text{Unit Contribution} \times \text{no of units} = \text{Total Contribution}$$

$$£10.00 \times 5,000 = £50,000$$

Now calculate Net Profit at 5,000 units

$$\text{Total Contribution} - \text{Fixed Costs} = \text{Net Profit}$$

$$£50,000 - £24,000 = £26,000$$

## Margin of safety (MOS)

- $\text{MOS} = \text{Budgeted Sales} - \text{Break even point} * \text{CPU}$
  - $= (5000 - 2400) * 10.00$
  - $= \text{Rs.}26,000.00$
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# Important Formulas



$$\text{Total Contribution} = \text{TR} - \text{VC}$$

$$\text{Unit Contribution} = P - \text{AVC}$$

$$Q^* = \text{FC} / (P - \text{AVC})$$

$$\text{BES} = \text{FC} * \text{TR} / (\text{Total Contribution})$$

$$Q_t = (\text{FC} + \pi_t) / \text{CM}$$

$$\text{P/V Ratio} = 1 - \text{VC} / \text{Sales}$$

$$\text{MOS} = \text{Budgeted Sales} - \text{Break even point} * \text{CPU}$$